SUPERLUMINOUS SUPERNOVAE

THE GORGEOUS MEGA-VILLAINS OF OUTER SPACE

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Beauty, as they say, depends on how you perceive it. The vast space has eons of mysteries tucked away in distant corners, much of which we have not yet set sight upon. However, our greatest scientific and engineering efforts are reaching out to see the farthest, darkest corners of outer space with greater panache than ever before. In this instance, what we witnessed at the beginning of 2016 was something that scientists and researchers are stating was a 'hypernova', or a superluminous supernova.

For reference, a supernova itself is a sight to behold. It is the sharp, violent death of a star when it finally runs out of its fuel, emitting immense amounts of energy and light and shining its brightest, before eventually fading and burning out. Yet another type of supernova occurs when two white stars, bigger than our dwarf star the Sun, collide with each other, leading to the instability of their cores and a subsequent thermonuclear reaction. These, as held by researchers, have a characteristic type of luminance, hence forming a metric in the scaling of supernovae.

The latest type, however, is rarer, and incomprehensibly more hostile. Superluminous supernovae, as the name itself suggests, is something that is far more violent and hostile than the average death of a star. These are brighter than the brightest, compounding immense amounts of force and energy at the point of explosion. They also occur across multiple stages, and the overall phenomena has only recently come to light.

The most beautiful moments are also the most violent

That itself is not really surprising, seeing how the first superluminous supernovae are estimated to be only about a few billion light years away from us. What is it, that makes such gargantuan creations in outer space? How are they formed, and what are the forces that are at play there? The research has only about begun, but from what has been found, superluminous supernovae are as mesmerising as their very name.

What is a superluminous supernova?

Quite literally, a superluminous supernova is the explosion of a star that is multiple times brighter and more violent than that of a standard explosion. These are typically about 100 times brighter than a supernova, and from what researchers have revealed so far, occur mostly because of exceptional circumstances - collision of giant stars or exceptional thermonuclear explosions being the more 'common' of exceptional cases.

Although we have observed a number of stellar deaths leading to this phenomenon, the entire range of causal factors is not yet known. Out of the four broad categories into which supernovae are classified, researchers have at least experienced the corresponding superluminous phenomenon of three types. This relatively new field of research